Consistency and Stability of XAI Explanations to Understand QSAR Models trained for Predicting Developmental Toxicity.

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1. INTRODUCTION

QSAR models based on Machine Learning predict molecular toxicity from numerical molecular descriptors [1]. However, different algorithms may learn different patterns due to their inductive biases, leading to distinct model behaviors.

Explainable AI (XAI) methods, such as LIME and SHAP [2,3], help identify the molecular features contributing to model predictions. However, the reliability of these explanations depends on their consistency and stability, which must be assessed before using them to interpret QSAR models.

How consistent and stable are SHAP and LIME explanations across QSAR models with different inductive biases

2. METHODOLOGY

I Datasets [4]

Dataser (D1): 292 comp.

Dataset (D2): 527 comp. (90% tóxicos [4])

II Features

27 RDKit Descriptors

III QSPR Models

DSVM (RBF) | **NB** | **KNN**

IV Explainers

SHAP Kernel-SHAP | **LIME** 5000 sapmles

V Assessment [5]

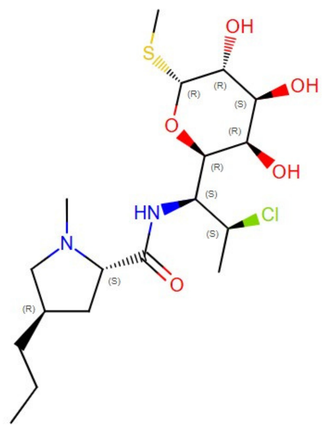
RC: Rank consistency
IS: Importance stability

GES: ½ (RC + IS)

3. RESULTS AND DISCUSSION

Case study: Molecule 14 - Clindamycin

In clindamycin, toxic alert structures (toxicophores) coexist with detoxifying groups (OH, sugar, thioether) that increase hydrophilicity and detoxification.



D1

True label: NON- TOXIC

SVM NON-TOXIC

Naive Bayes NON-TOXIC

KNN TOXIC

N-dealkylation

Heterocyclic fragments

Detoxification-related feature

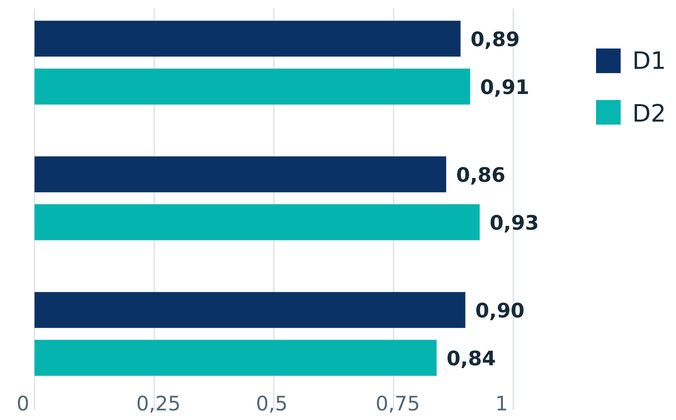
Different inductive biases, different molecular cues

SVM emphasizes selected molecular patterns; **NB** combines fragment-level evidence; **KNN** relies on local similarity in descriptor space.

Table 2. Stability metrics on D1 and D2 for SVM, NB, and KNN, using SHAP and LIME.

		RC		IS	
Model	**Method**	**D1**	**D2**	**D1**	**D2**
SVM	SHAP	0,7761	0,7482	0,4144	0,4973
SVM	**LIME**	**0,8778**	**0,8690**	**0,8513**	**0,9970**
NB	**SHAP**	**0,8118**	**0,8772**	**0,9768**	**0,9372**
NB	LIME	0,6889	0,6778	0,3704	0,3852
KNN	SHAP	0,7493	0,7597	0,3442	0,5220
KNN	**LIME**	**0,8324**	**0,8352**	**0,9635**	**0,8533**

Figure 1. Global Explanation Score (GES) of highlighted combinations



The most suitable explainer depends on the QSAR model: LIME performs better for SVM and KNN, while SHAP is more consistent and stable for Naive Bayes.

4. CONCLUSIONS

Explanation consistency and stability depend on the underlying QSAR model. LIME performs better for SVM and KNN, while SHAP achieves higher scores for Naive Bayes, with this pattern replicated across both datasets. Moreover, feature-importance rankings were generally more consistent across datasets than importance values, supporting the use of complementary metrics to assess explanation reliability. **These findings highlight the importance of evaluating XAI methods in relation to the models they explain, rather than assuming a universally superior explainer.**

References:

[1] Ponzoni, I., et al. Wiley Interdiscip Rev Comput Mol Sci, 2023, 13(6), e1681.
[2] Ribeiro, M.T., et al. Proceedings of KDD, 2016, 1135–1144.
[3] Lundberg, S.; Lee, S. Adv. in Neural Information Processing Systems, 2017, 30.
[4] Togo, M.V., et al. J Chem Inf Model, 2023, 63(1), 56-66.
[5] Munoz, C., da Costa, K., Modenesi, B., & Koshiyama, A. (2023). Evaluating explainability in machine learning predictions through explainer-agnostic metrics. arXiv preprint arXiv:2302.12094.